MIL-C-3098/49B
30 July 1974
SUPERSEDING
MIL-C-3098/49A
22 September 1964

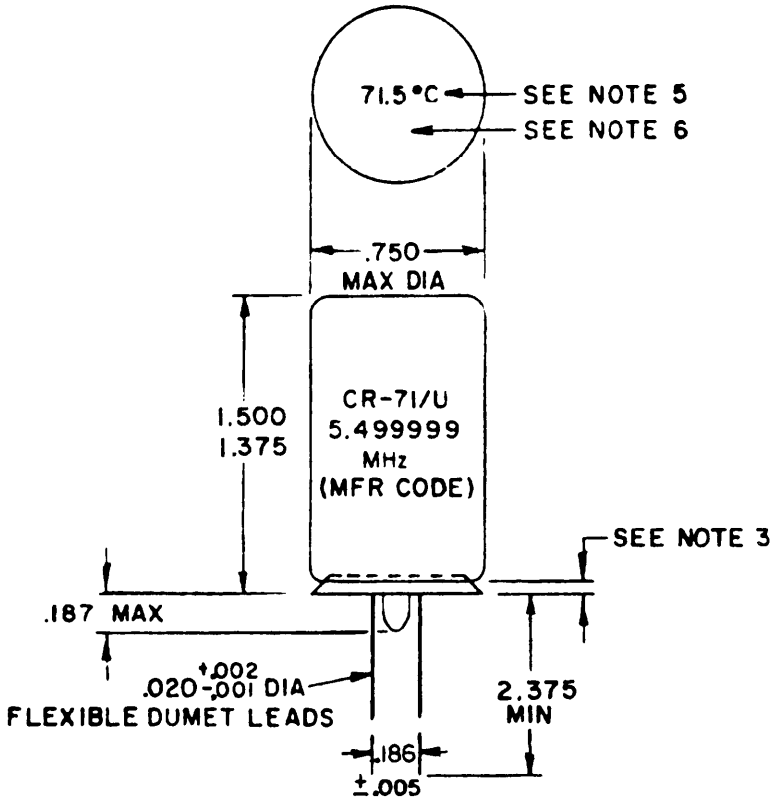
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-71/U

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of Specification MIL-C-3098.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

Pertinent characteristics: 4.5 to 5.5 MHz; fifth overtone; controlled; antiresonance



INCHES	MM
.001	.03
.002	.05
.005	.13
.020	.51
.186	4.72
.187	4.75
.750	19.05
1.375	34.93
1.500	38.14
2.375	60.33

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. No limiting dimensions for seal area.
4. Leads shall be straightened and tinned.
5. Reference temperature, point of zero temperature coefficient to one decimal point.
6. Serial number, if assigned, shall be marked on the crown of the glass envelope.

(B) denotes changes

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-30/U (MIL-H-10056/12).
Dimensions, marking, and configuration	See figure.
Frequency range	4.5 to 5.5 MHz, inclusive.
Capacitance, shunt	4 ±0.4 pF.
Frequency tolerance	±0.00008 percent.
Frequency stability	±0.00008 percent.
Equivalent resistance	175 ohms, maximum 100 ohms, minimum
Antiresonance, load capacitance	32 ±0.5 pF.
Mode of oscillation	Fifth mechanical overtone.
Reference temperature 1/	65° to 77°C.
Temperature range:	
Operating	±0.5°C from reference temperature.
Government reference standard test set	AN/TSM-17.
Drive level, rated 2/	70 μA ±14 μA.
Ratio of capacitances 3/	31,000 to 38,500, inclusive.
Shock:	
Frequency change permitted	±0.00001 percent.
Equivalent resistance change permitted	±10 percent.
Vibration, high frequency: 4/	
Frequency change permitted	±0.00001 percent.
Equivalent resistance change permitted	±10 percent.
Orientation: 5/	
Frequency change permitted	±0.000001 percent.
Aging (frequency change permitted): 6/	
Qualification	5 parts in 10^9.
Quality conformance	1.5 parts in 10^9.

1/ The zero temperature coefficient of the crystal unit shall be determined using the test set, and adjusted for operation at rated drive level, as follows:

(a) Crystal unit temperature shall be set at 80°C for 2 hours.
(b) Crystal unit temperature shall be lowered at a rate not to exceed 1°C each 5-minute period over a temperature range from +80 to +60°C.

(1) Frequency and temperature measurements shall be recorded either continuously or at intervals not to exceed 1 second.
(2) The temperature at which frequency is minimum (which is the zero coefficient of frequency with respect to temperature) shall be designated as the reference temperature of the unit. It shall occur between the temperatures of +65° to +77°C. The referenced temperature shall be determined to the nearest 0.1 °C and marked on the crystal unit in accordance with the figure.

2/ The drive level shall be varied from the 70-microampere (μA) level in steps of 1 dB to cover the range of at least 40 to 90 μA, inclusive, while the temperature is maintained constant within ±0.01°C of the initial temperature. The frequency shall be measured at each drive level step. The frequency change due to drive-level variation shall be not greater than ±2 parts in 10^9 per 1 dB current change.

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3/ The frequency of parallel resonance with external load of 32 pF, and also the frequency of series resonance, shall be measured with the crystal unit at room temperature (25° ±5°C). The ratio of capacitance shall be calculated by the following formula:

$$r = \frac{C_O f_S}{2 \Delta f (C_O + 32)}$$

Where: r = ratio of capacitances.
C_O = measured static capacitance, in pF, to one decimal place.
f_S = series resonant frequency, in Hz.
Δf = difference between parallel and series-resonant frequencies.

4/ Method 204 of MIL-STD-202, test condition A, except upper frequency limit is 150 Hz. Measurements after test shall be made at a temperature within ±0.1° of temperature before test.

5/ The crystal unit shall be oriented so that the plane of the resonator is vertical. The unit shall be slowly rotated 360° in the plane of the resonator. During the rotation; the resonator shall be variously oriented in a vertical attitude and at 45° either side of the vertical. The frequency shall be measured continuously, with the temperature maintained within ±0.01°C throughout.

6/ Aging:

(a) Qualification inspection. Crystal units shall be oscillated within the operating temperature range for a period of 21 days. Frequency measurements shall be made starting the 12th day and made either continuously or once per working day through the 21st day. The difference in temperature between the initial measurement and any subsequent measurement shall not exceed 0.01°C.

(b) Quality conformance inspection. Crystal units shall be maintained at a temperature of 71° ±6°C for a period of 48 hours. The units shall then be maintained within the operating temperature range for a period of 4 hours. The frequency shall be measured continuously during this period.

NOTES: This crystal-unit type is to be used for special applications and for replacement
1. purposes only. Specific approval of the Government is required prior to each intended use (see MIL-STD-683).

2. NATO preferred type NXT32.

Custodians:
Army - EL
Navy - EC
Air Force - 84

Review activities:
Army - EL, MI, MU, SM, SL
Navy - EC
Air Force - 11, 80
DSA - ES

User activities:
Army -
Navy - AS, CG, MC, SH
Air Force - 17, 19, 84

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-0360-19)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-3098/49B CRYSTAL UNITS, QUARTZ, CR-71/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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